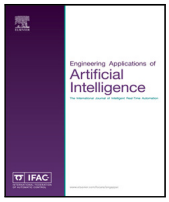
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Research paper

Fuzzy rules intelligent car real-time diagnostic system



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ABSTRACT

Advanced collaborative technologies for Artificial Intelligence of Things (AIoT) play main role in the development of smart technologies. In this article we present research results on the development of novel complex type-2 fuzzy control system to help in real-time measuring and diagnostic of vibration, temperature and air quality for optimal car driving. Real-time information allows the driver to be informed about the possibility of a potential failure. It gives a chance to avoid failure and related consequences. The paper presents both theoretical aspects of applied type-2 fuzzy control and practical constructions for sensors, electronics and software implementations to compose a novel control module implemented for car radio systems. The developed system was tested in various driving conditions and results of measurements and predictions were confirmed by the certified car diagnostic station.

1. Introduction

The rapid development of artificial intelligence methods and IoT infrastructure based modules on widely available sensors and electronic modules enables construction of measurement and diagnostic systems that facilitate operation of advanced devices. In currently produced vehicles, self-monitoring systems for most of devices in the vehicle are installed on a large scale. For a long time, cars have been equipped with systems that carry out diagnostics during the use of the vehicle. This diagnosis happens in real time on board of the vehicle, such as self-diagnosis of injection system, anti-skid system and safety systems (airbags and curtains). Systems based on artificial intelligence, produced to support the driver, are solutions that enable, for example, autonomous parking of the vehicle. The paper (Alyas et al., 2019) proposes the use of Mamdani Fuzzy Inference System (MFIS) to facilitate the process of parking. Fuzzy models are suitable for this type of problems, due to the relatively uncomplicated possibility of tuning. In advanced control systems we can often find a variety of IoT applications, which enable fast control and data transmission. In Liazid et al. (2023) was discussed how to reduce transmission requirements by data aggregation techniques for IoT wireless sensor networks. Such solutions are also reported in car industry. In Auer et al. (2022) was presented an IoT model for car sharing management in which mobility of cars was under permanent control by using IoT infrastructure.

The infrastructure installed in motor vehicles allows for the diagnosis of surface quality and, because of the connection to the Internet, allows for marking places where the road surface has an anomaly (Woźniak et al., 2022). The article presents a diagnostic method based on type-2 fuzzy inference. The specificity of the measurements detected by the sensors was identified by the fuzzy system. The results were sent via GSM network to the server application, where the anomalies detected on the road surface were marked in geolocation system.

Fuzzy sets of the second type are a tool that takes into account measurement uncertainty very well. The tool which is using fuzzy logic due to uncertainties was used to determine the vehicle path by autonomous cars, as presented in Yang et al. (2021). In autonomous vehicles, type-2 fuzzy sets are used to reduce energy consumption in a hybrid solution (Phan et al., 2020).

Fuzzy sets are also used in problem of finding optimal routes and selecting right vehicles for these routes in terms of energy consumption. This problem was raised in Mehlawat et al. (2019). Recently, fuzzy sets have also been used in construction of similarity measures used in classification process, which was presented in Cherif et al. (2021). Fuzzy sets of the second type were also used to create a clustering algorithm, as described in Yang et al. (2020).

We can find many interesting ideas for the composition of adaptive control models, where as adaptive control different approaches were proposed in variety of applications. The adaptivity to the input features was composed by using different type of techniques. In Shi et al. (2023) was presented a model of steer-by-wire control system with an adaptive observer model. Fuzzy wavelet type adaptive control model for aerial vehicles was presented in Song et al. (2024), while in Stojanović (2023) was proposed a construction of control model for hydraulic type

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Fig. 1. Research model: Cadillac Seville year 1991.

actuator. Fuzzy control was also proposed for nonlinear systems (Sun et al., 2023).

Control models in various type are used for real objects modeling. This type of systems make possible evaluation of data in representing operation of complex technical systems. In Feng et al. (2023b) was proposed a model for gear surface degradation, while in Ni et al. (2023) rolling element bearing diagnostic module was defined by using computational intelligence approaches. It is also very common to work with different type series of data. In Feng et al. (2023a) was proposed to implement an intelligent control module for surface wear progression. The model presented in Zhuang et al. (2022) was developed for linear systems with nonuniform trial which are often used in automation control modeling. In Djordjevic et al. (2023) data-driven control was developed for hydraulic servo actuator, while fault-tolerant control for the same type of technical appliance was proposed in Stojanović (2023). One can also find approaches that adapt the control scheme during changing time intervals. This type model for unmanned surface vehicle was developed in Song et al. (2023). In our work, a devoted measurement infrastructure was created based on popular sensors used in IoT solutions and wireless communication (WiFi) was used between installed sensors connected to an application running on a Raspberry Pi computer. Thanks to our solution, the need to create wire connections difficult to make in environment of an existing car (not at the stage of its construction) was avoided. The proposed diagnostic system is designed to preventively warn the user of a potential fault. For this purpose, a fuzzy system of the second type was created, which, based on ongoing measurements and expert knowledge, warns about emerging irregularities that the driver is not able to notice. This type of solution is a recommended tool, especially for older vehicles. The vehicle used as our research model is shown in Fig. 1, where obtaining spare parts is very difficult or even impossible. A particularly important aspect of diagnostics is determining the operation of the drive system based on temperature measurements in selected places of cooling system. The second important aspect is mechanical damage to the chassis, which is diagnosed based on vibration measurements and harmonic spectrum analysis. The work includes practical examples of fault detection in monitored objects.

Key contributions of the paper:

- New rule-based measurement methods for oldtimer cars,
- Novel use of type-2 fuzzy control to model sensor reading which have uncertainty,
- Real-time diagnostic system.

Major findings of the research:

- Real-time diagnosis of shock absorbers based on proposed adaptive vibration analysis model, confirmed at the certified diagnostic station,
- Devoted models of electronic systems for an oldtimer car.

2. Complex AIoT system model

Using the vehicle causes wear of the drive and braking system components, which may lead to the appearance of additional vibrations. These vibrations cause accelerated wear of other elements of the running gear. For example, a worn drive-shaft joint will cause additional vibrations on one side of the vehicle driveline, which may be dampened by the vehicle's suspension system. Even if vibrations may be felt by the driver, sometimes it will be impossible to locate their source, ie. a source of vibration could be an unevenly worn or deformed brake disc. In this case, vibrations will appear mainly when the car is braked. In all situations, it is important to detect vibration abnormalities early before they can be clearly felt by the driver and before they influence elements of the car. It should be emphasized here that the driver given a slowly increasing stimulus gradually gets used to them, which affects the delay in detecting irregularities by his senses. Therefor a complex AIoT system may help in real-time diagnostic. In implemented solution, we have also placed accelerometers. Sensors were mounted on the hubs of all wheels of the drive system as shown in Fig. 3 to help in real-time prediction of vibration.

Another aspect to consider is driveline temperature monitoring. Primary, the parameter is the temperature of the engine coolant. It's temperature is influenced by many factors, including engine load, driving speed and ambient temperature, indirectly also influenced by the temperature of the gearbox oil, which is cooled in the same cooler. The increase in the temperature of the gearbox oil depends on the load and the operating mode (torque converter operation). We can also monitor temperatures in other parts of the cooling system. Sensors located in our system have been marked on the diagram of the cooling system shown in Fig. 4. Analyzing the operation of the drive system, it was decided to install temperature sensors determining the temperature of the cooling liquid: in the engine T_1, at the entrance to the cooler T_2, and at the exit from the cooler T_3. Additionally, in the proposed system, engine oil temperature sensors were installed at the entrance to the cooler T_4 and at the exit T_5, the temperature of the gearbox oil at the entrance T_6 and at the exit T_7. Proposed solution can be used in any car in which the engine oil is fed to the heat exchanger and similarly the gearbox oil (classic automatic gearbox cooperating with a torque converter). Due to the fact that the proposed system does not interfere with the design of the original vehicle components, all sensors of the tested temperatures were installed externally on the conductors of the tested medium. These sensors do not come into direct contact with the tested liquid (no interference with the structure). These sensors measure the temperature of the liquid through metal elements that conduct heat well. In addition, the temperature sensors have been thermally insulated from rolling away, minimizing the impact of external conditions on their measurements.

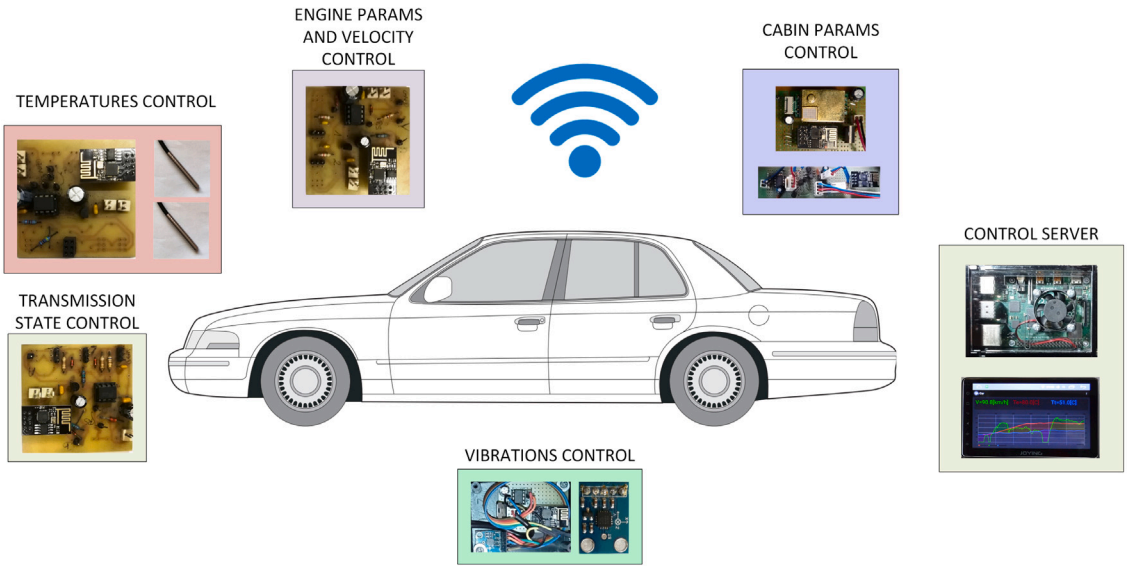


Fig. 2. Developed Artificial Intelligence of Things (AIoT) real-time car diagnostic system modules.

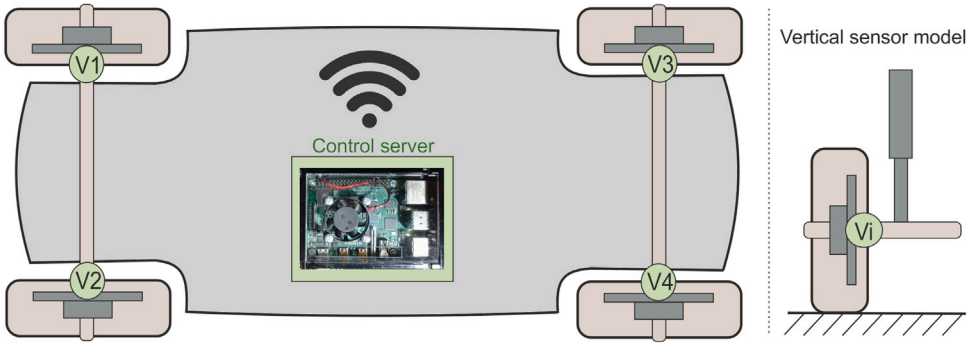


Fig. 3. Accelerometers in the car suspension system.

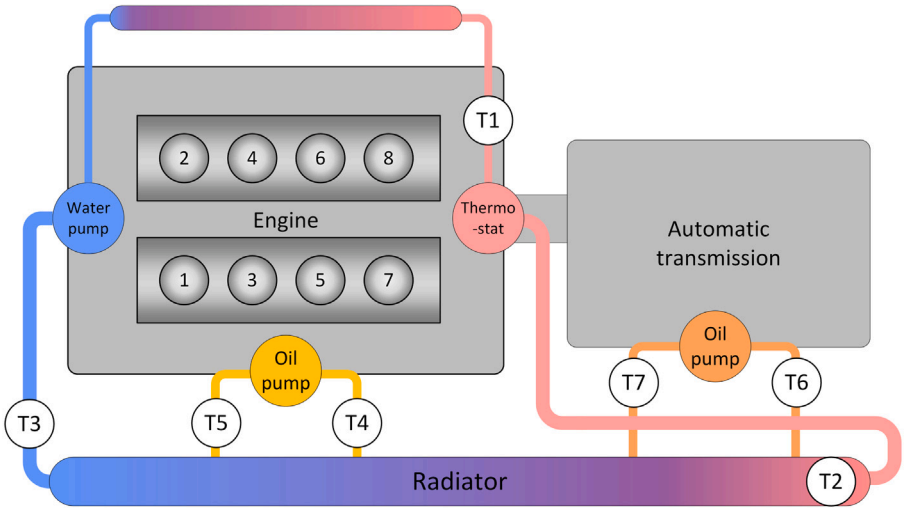


Fig. 4. Location of temperature sensors in the cooling system of the engine and gearbox applied in our fuzzy control model.

In addition, the system was expanded with a comfort module in the vehicle cabin. For this purpose, a CO_2 concentration measurement sensor and a temperature sensor as well as an executive system responsible for ventilating the vehicle cabin were used. The concept of the developed AIoT system is presented in Fig. 2.

3. IoT connection model

In order to carry out the tasks described above, an IoT system installed in the car was created. The master unit of which is a Raspbbery Pi class computer on which the server application is running. The

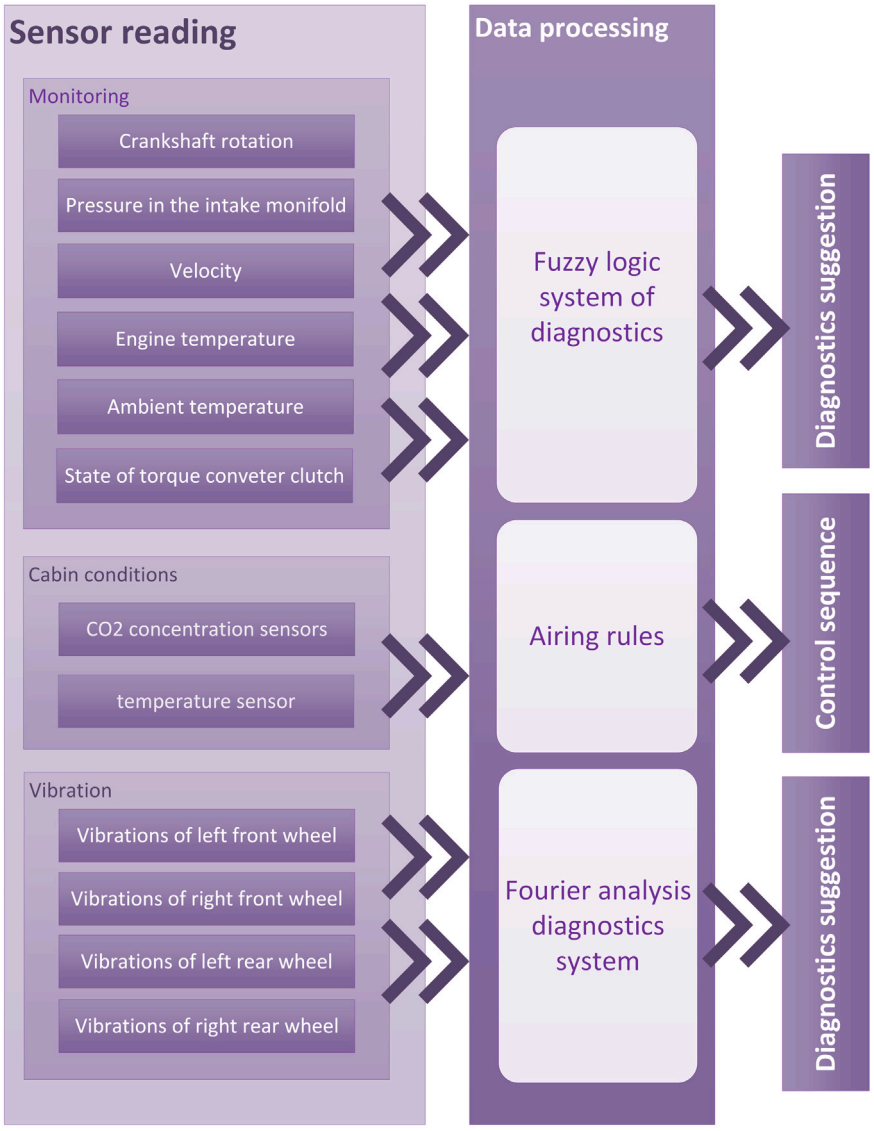


Fig. 5. Data flow between installed IoT elements.

sensors installed in the car communicate with the master unit via Wi-Fi network broadcast in the car. Communication between the measuring modules and the server takes place via Rest Api. The data received by the server application are then collected and stored in a local data base (SQLite) and processed by the proposed main diagnostic fuzzy control algorithm which analyzes measurements in terms of detecting possible irregularities. The client application of the system has been implemented on car radio mobile devices for Android. The operation of the system is visualized for the driver in real-time through the radio display, which communicates with the master unit also via Wi-Fi. The diagram of data flow in the composed IoT system is shown in Fig. 5. Let us now present electronic infrastructure developed for the tested car.

3.1. Developed electronic components

3.1.1. System for measuring basic parameters of the drive engine

Fig. 6 shows a schematic diagram of the measuring system used to record operating parameters. The task of the system is to measure: engine speed, linear speed of the vehicle, throttle opening and pressure in the engine intake manifold. In addition to the measurements, this system is responsible for transmitting them to the central unit of the system via Wi-Fi.

Measurements of all parameters are based on the ATtiny 13 microcontroller. The engine speed is measured based on the signal from the ignition module. Additionally, a system was used to protect the microcontroller against overvoltages that may appear in the recorded signal. Speed measurement is made on the basis of a signal obtained from the vehicle speed sensor. The signal reaching microcontroller is a digital signal. The measurement is carried out by counting the logic change. In addition, the microcontroller, using analog-to-digital converters, measures the throttle opening and the negative pressure in the engine intake manifold. The throttle opening is determined by measuring the signal from a sensor fitted as standard in the engine control system - the throttle opening sensor. Similarly, the measurement of the pressure in the intake manifold is based on the signal from the absolute pressure sensor in the intake manifold. The ATtiny 13 microcontroller is powered with 5 V through the voltage stabilization system IC1 with the attached output capacitors C3 and C4. Periodically, the ATtiny 13 microcontroller sends the recorded data to the ESP8266 system responsible for data transmission via Wi-Fi to the central unit of the system. Data transmission between the ATtiny 13 microcontroller and the ESP8266 system takes place via the RS232 bus at the TTL signal level. It should be emphasized that the maximum allowable supply voltage of the ESP8266 system is 3.3 V. For this reason, a second

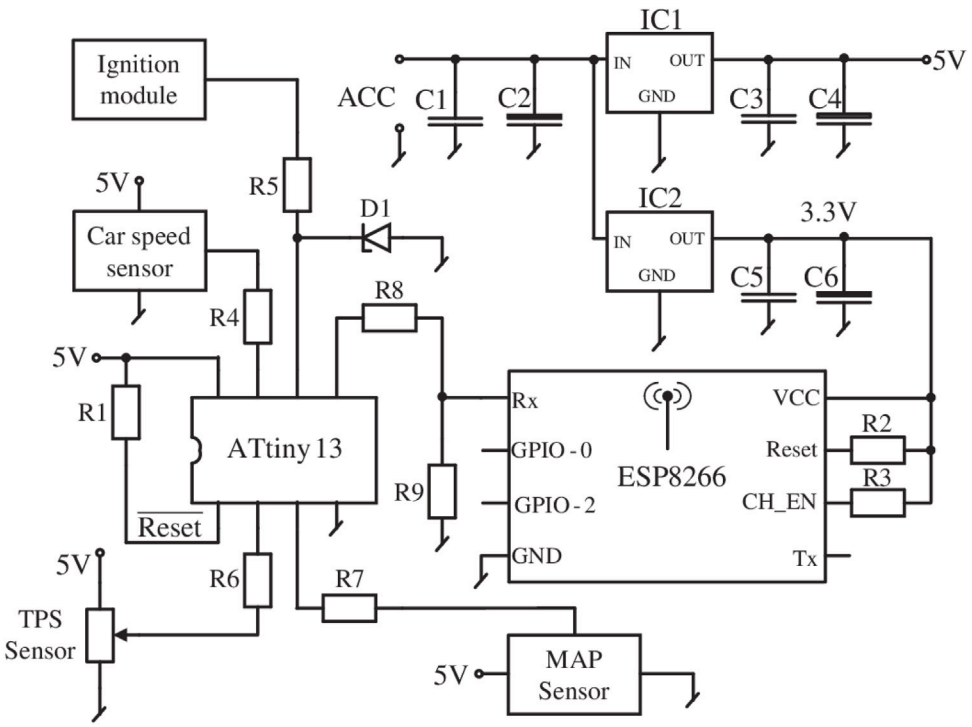


Fig. 6. Schematic diagram of the measuring system of parameters of the drive engine: engine speed, linear speed of the vehicle, throttle opening and pressure in the engine intake manifold.

voltage stabilizer IC2 was installed, cooperating with the capacitors C5 and C6, which give a voltage of 3.3 V at the output. In addition, to adjust the required voltage levels between these systems, a divider made of R8 and R9 resistors was used.

3.1.2. Gearbox parameter measurement system

Fig. 7 shows the measuring system used to record signals from a four-speed fully automatic gearbox. The system was built as before based on the Attiny 13 microcontroller and WiFi transmission based on the ESP8266 system. The power supply and level matching of RS232 transmission signal is the same as before. The microprocessor records the signal delivered from the PCM to the gearbox. This measurement is made at the gearbox coupling. On the basis of these signals, it is possible to determine currently selected gear from 1 to 4 (signals from solenoids "A" and "B") given through pins F and G of the gearbox connector, and obtain information whether the drive train uses a torque converter or it works with the use of a visco-kinetic clutch (signal from the terminals of the TCC/VCC solenoid), changed via pins A and D of the gearbox connector. These signals are digital signals supplied to the pins of the microcontroller through a system of resistors (R4 - R7) and zener diodes (D1 - D4) limiting the maximum voltage supplied to the input of the microcontroller. In addition, through the analog-digital converter, the microcontroller can measure the average value of the PWM signal. PCM system installed in the vehicle has ability to adjust the slip of the Visco-kinetic clutch. The PWM signal is supplied to the analog-to-digital converter through a low-pass filter system made of the resistor R1 and the capacitor C7 in order to enable measurement of the average value of the PWM signal.

3.1.3. Temperature recording system of the drive engine and cooling system

Fig. 8 shows a diagram of the system for measuring and recording temperatures of the drive system in selected places marked in the diagram shown in Fig. 4. The measurement system is based on the ATtiny 13 microcontroller and the ESP8266 Wi-Fi transmission system. DS18B20 systems were used as temperature sensors, these are Dallas Maxim Integrated temperature sensors transmitting data via 1-wire bus. Thanks to this solution, all sensors used in this system are connected to one pin of the ATtiny 13 system. Temperature measurement is carried out in such a way that an order to perform temperature measurement is sent to all sensors, then the microcontroller using unique sensor numbers (DS18B20) polls each of the sensors o the measurement result. This solution allows to assign a sensor to a selected temperature measurement point.

3.1.4. System for detecting the state of the drive transmission components and suspension system

This diagnostics were carried out based on the use of accelerometers, which were used to measure vibrations on selected parts of the vehicle. The recorded vibrations are subjected to an appropriate analysis and interpretation based on the mathematical apparatus of the Fourier transform analysis. In our solution, accelerometers were mounted on the hubs of all wheels of the drive system as shown in Fig. 3. The scheme of the implementation of a single measurement system based on the GY-61 accelerometer module is shown in Fig. 9. A commonly available DXL335 accelerometer measuring direction of three perpendicular vectors was used for the measurement. Accelerometers were installed in such way that the direction of travel coincides with the axis marked X on the accelerometer and the Z axis is perpendicular to the ground. At the output of the accelerometer there are three analog signals corresponding to the overload in the direction of each axis of the XYZ system. Accelerometer output signals are delivered to three analog-to-digital converters of the ATtiny 13 microcontroller, which forward them in packets to the Wi-Fi ESP8266 data transmission system.

3.1.5. Cabin carbon dioxide level measurement system

Fig. 11 presents system of the ESP8266 WiFi transmission system, the DS18B20 temperature sensor and the CO_2 concentration measurement system (MH-Z19B). The ESP8266 chip is periodically interrogating the temperature sensor and the CO_2 concentration sensor. After reading the measurements, it transfers them to the central unit. The model developed in Fig. 11 can be used in systems where the internal

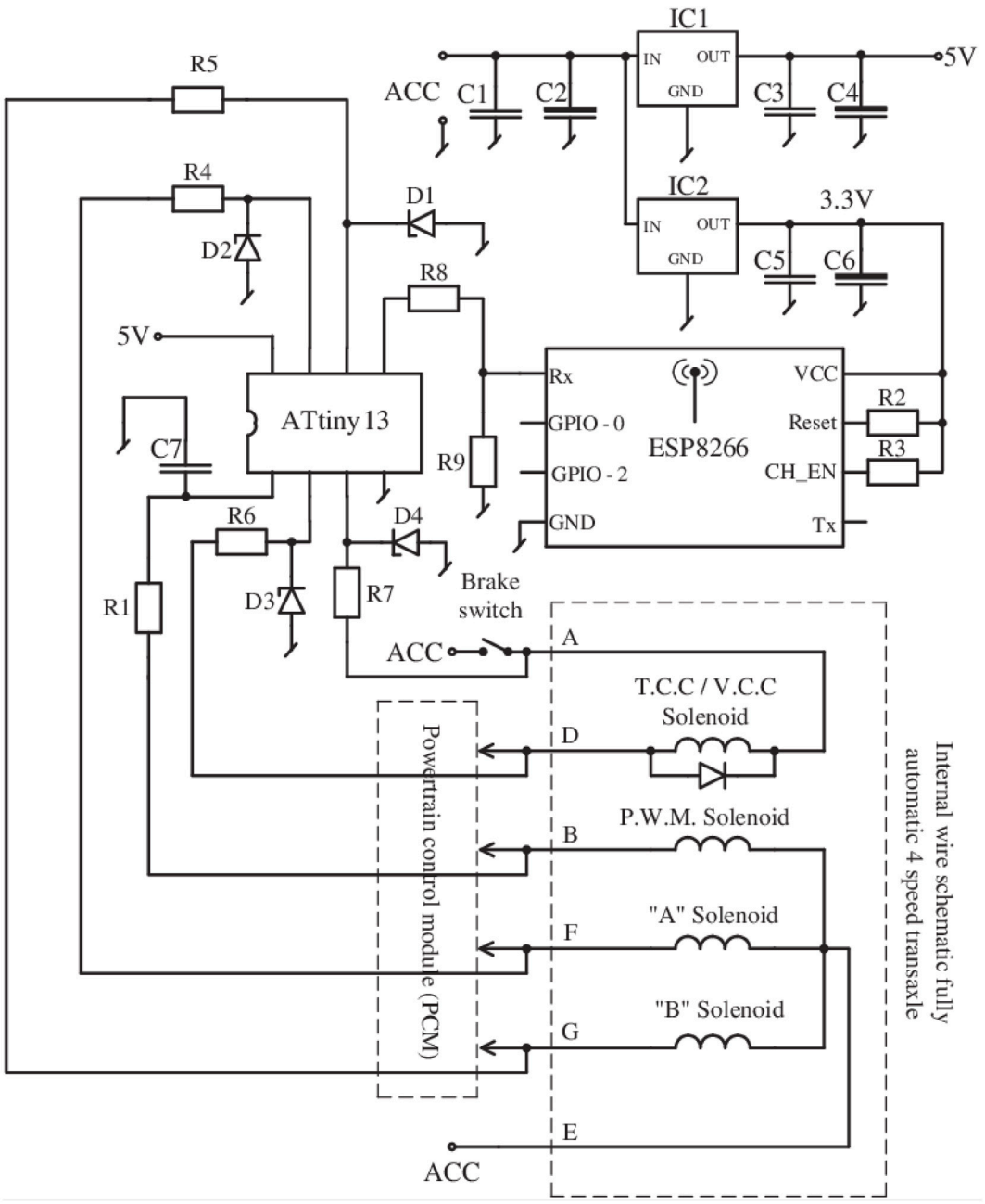


Fig. 7. Schematic diagram of the signal measuring system of a 4-speed fully automatic gearbox.

to external circulation takes place through a solenoid valve. Due to the fact that the ESP8266 system can be supplied with a maximum voltage of 3.3 V, the solenoid valve changing the internal to external circuit is controlled by the system of two transistors. The ESP8266 system controls the BJT transistor (T1), a voltage of 3.3 V is sufficient for this, while the vehicle installation voltage is used to fully open the T2 transistor.

4. Complex artificial intelligence model

Both in modern and classic cars, the diagnostic system informs the driver only about exceeding the permissible maximum temperature of the engine, regardless of the operating mode and load. This approach to diagnostics does not allow for the observation of wear of certain components. However we can non-invasively monitor the temperatures at various measurement points of the drive system, which will allow for earlier detection of certain abnormalities and prevent damage. For the interpretation of the measurement results, a fuzzy inference system type-2 was developed, which was based on the measurements of the engine temperature, ambient temperatures, vehicle speed and the determination of the engine load during its operation and the operation of the transmission system (TCC/VCC). The system is based on fuzzy inference, because it is possible to define all the rules corresponding to the correct and disturbed operation of the tested system and the emergency state by expert knowledge. In addition, type-2 fuzzy sets well model the uncertainty resulting from both the errors that the measurements are loaded with and the exact location of the sensors built into our system, which are mounted on the outer wall of the

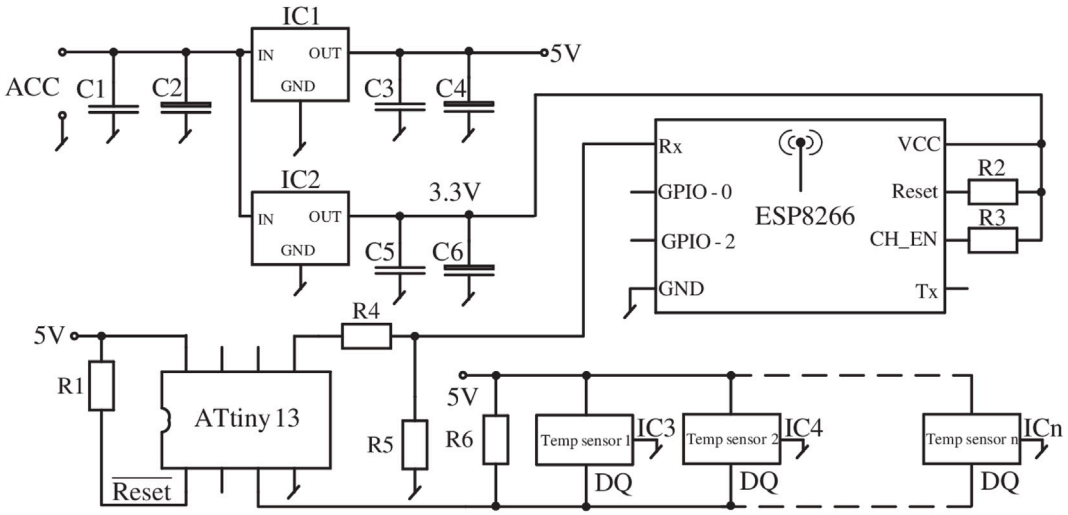


Fig. 8. Schematic diagram of the temperature measuring system of the drive engine and gearbox cooling system.

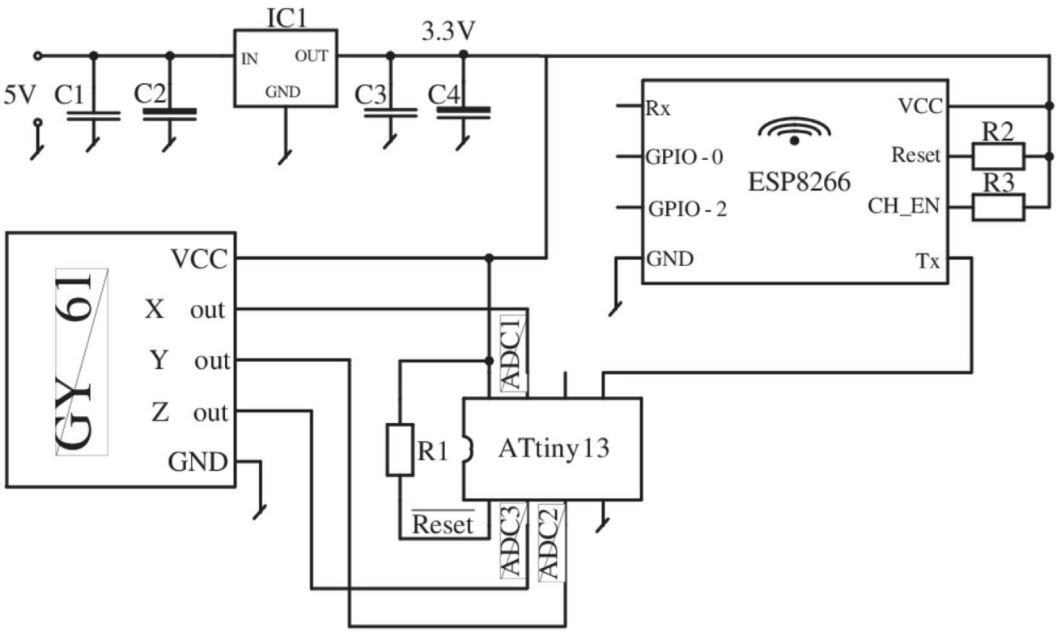


Fig. 9. Schematic diagram of the vibration measurement system.

cooling liquid pipe. Before we proceed to a detailed description of the fuzzy reasoning system, we will present the theoretical background to better understand type-2 fuzzy systems.

4.1. Type-2 fuzzy logic

At this point, we will focus on presenting the basic concepts of type-2 fuzzy sets. First we give a definition of fuzzy set type-2. *Fuzzy set type-2*, is $\tilde{A} \subseteq X$, for X which is a decision space we name the set of ordered pairs:

$$\left\{x, \mu_{\tilde{A}}(x)\right\}, \tag{1}$$

where the first element of this pair x is an element of the fuzzy set, and the second element of the pair is a function of membership $\mu_{\tilde{A}}(x)$ of the element x to the fuzzy set $\tilde{A}$ which is the fuzzy set type-1 belonging to $J_x \subset [0,1]$ what we mark:

$$\mu_{\tilde{A}}(x) = \int_{u \in J_x} f_x(u)/u. \tag{2}$$

Let us now define the trace of uncertainty.

Let $J_x \subset [0,1]$ denotes the basic affiliation of x. The trace of uncertainty SN of a fuzzy set type-2 $\tilde{A} \subseteq X$, is a limited area composed of all points of the basic affiliation of x elements:

$$SN(\tilde{A}) = \bigcup_{x \in X} J_x. \tag{3}$$

In the implementation of a fuzzy system, the trace of uncertainty determines the range of calculations for the type-2 fuzzy set under consideration.

Fig. 12 shows an example of a type-2 fuzzy set. The top plot represents a type-2 fuzzy set, where the trace of the set is marked in blue. In addition, one point x_0 was distinguished and for it $J_{x_0} = [y_1, y_2]$ was marked in red, which is the domain of the uncertainty function of the second type. The membership of an element in a set is defined by a fuzzy set of type-1. Two examples of membership functions for the set of the first type are shown in the graphs below. The chart on the left shows an interval function, i.e. constant in the domain J_{x_0}. Then we are dealing with an interval fuzzy set of type-2. The chart on the right shows the Gaussian membership function in red.

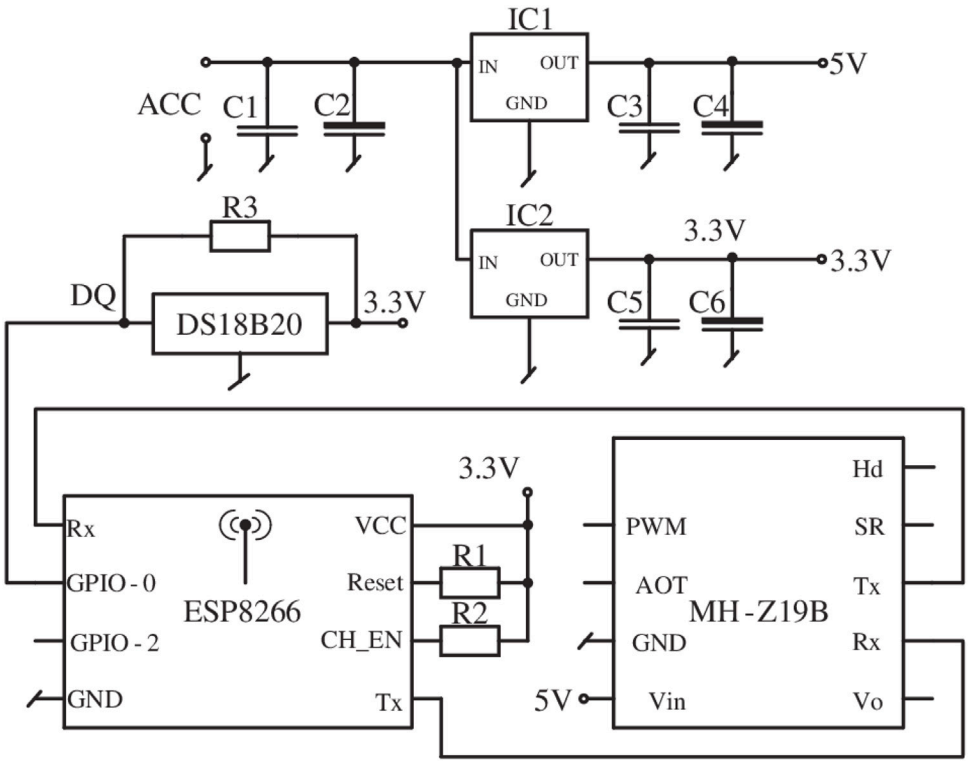


Fig. 10. Schematic diagram of the car cabin carbon dioxide condition measuring system.

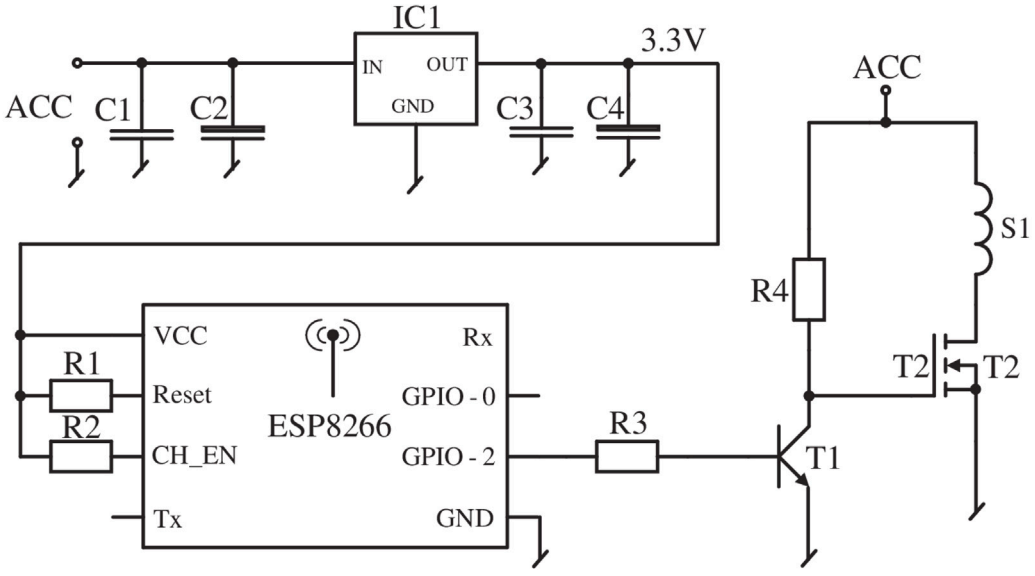


Fig. 11. Schematic diagram of the control system switching indoor air with outdoor air.

In our considerations, we assumed that we use interval type-2 fuzzy sets. The control system we propose assumes the fuzziness of all measurement quantities: engine speed, vehicle speed, engine temperature, ambient temperature, vacuum in the intake manifold. Additionally, we determine the torque, which we also treat as a fuzzy quantity. Thus building a fuzzy system of the second type, we must define the operation of the sum and products of fuzzy sets to adjust it to our assumptions.

Let sets $\tilde{A}$ i $\tilde{B}$ be defined:

$$\tilde{A} = \int_{x\in X}\left(\int_{u\in J_x^u} f_x(u)/u\right)/x, \tag{4}$$

$$\tilde{B} = \int_{x\in X}\left(\int_{v\in J_x^v} g_x(v)/v\right)/x. \tag{5}$$

Sum of sets $\tilde{A}$ i $\tilde{B}$ is a fuzzy set type-2 of membership function:

$$\mu_{\tilde{A}\cup\tilde{B}}(x) = \int_{u\in J_x^u}\int_{u\in J_x^v} f_x(u) \overset{T}{*} g_x(v)/u \overset{S}{*} v. \tag{6}$$

Intersection of sets $\tilde{A}$ i $\tilde{B}$ is a fuzzy set type-2 of membership function:

$$\mu_{\tilde{A}\cap\tilde{B}}(x) = \int_{u\in J_x^u}\int_{u\in J_x^v} f_x(u) \overset{T^*}{*} g_x(v)/u \overset{T}{*} v, \tag{7}$$

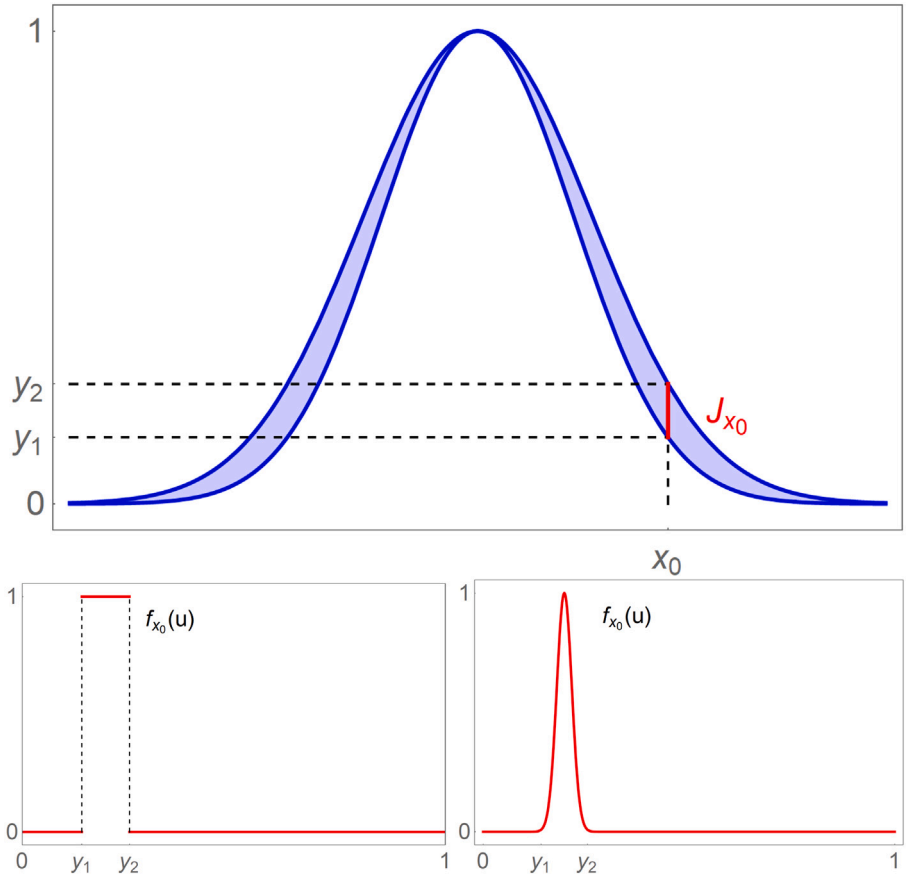


Fig. 12. An example of a fuzzy set of the second type (top) and two examples of a membership function for a selected point x_1 (bottom).

where T^* is t-norm 2nd type affiliation.

Our research presented in this paper assumes the use of interval type-2 fuzzy sets and we will use the minimum operation as $t-$norm and maximum operation as $s-$norm.

Fuzzy systems also require specifying a mechanism to determine the final result. For this purpose, the sharpening mechanism is used. i.e. it is necessary to obtain one specific value of the control based on the resulting type-2 fuzzy set. For the second type of fuzzy sets, the sharpening process is a two-stage process: the first step consists in sharpening the type-2 set and as a result, the first type set is obtained, and then in the second step, sharpening the first type set to a specific numerical value.

Assuming the discretization of the X set to the R set, the values $x_1, \ldots, x_R$, the fuzzy set of type-2 $\tilde{A}$ can be written:

$$\tilde{A} = \sum_{k=1}^{R} \left[\int_{u \in J_{x_k}} f_{x_k}(u)/u \right] / x_k. \tag{8}$$

The set resulting from the reduction of the type is called a centroid. For the discrete form of the set $\tilde{A}$, centroid is a set of type-1 and has the form:

$$C_{\tilde{A}} = \int_{\theta \in J_{x_1}} \cdots \int_{\theta \in J_{x_R}} [f_{x_1}(\theta) * \ldots * f_{x_R}(\theta)] \Big/ \frac{\sum_{k=1}^{R} x_k \theta_k}{\sum_{k=1}^{R} \theta_k}. \tag{9}$$

Guided by results of Torshizi et al. (2015) on the review of methods for reducing the type of fuzzy sets type-2 and taking into account the fuzzy system we are considering, we decided to use the algorithm proposed by Karnik–Mendel.

The basic and the most important assumption for the diagnostic system based on fuzzy inference is the detection of operating states of the drive system, which may indicate certain irregularities in its operation. The purpose of this system is to prevent a serious failure, which may lead to a deteriorating condition of the car.

The uncertainty of the data we used in our model depends on the accuracy of the measurement sensors used. All sensors used are presented in the article, and their accuracy can be read from the product data sheet. Uncertainties may also result from time shifts, but in our case they are not important due to the time constant of the entire system. However, by analyzing the read values, we can present the following explanations for the possible uncertainty of the proposed decision model given in Fig. 13:

- chart (a) presents possible engine operation ratios in relation to the engine shaft revolutions. Uncertainty means the possibility of adjusting the engine performance in particular engine speed ranges.
- chart (b) presents uncertainty of the incomplete measurement of engine operating vacuum depending on the load placed on the engine (i.e. how much fuel mixture it has in it to drive faster).
- chart (c) presents uncertainty of measuring the engine torque that was generated in relation to the shaft speed and the recorded vacuum in the intake manifold.
- chart (d) presents normal engine operation at 90 degrees. In other cases, uncertainties mean possible deviations of the engine operating temperature from the expected value. For example, “hot” means that the engine has lost power above max. temperature with some measurement accuracy reflected in uncertainty.
- chart (e) presents uncertainty in the analysis of the operation of the entire engine in relation to the ambient temperature, i.e. the temperature outside the vehicle.
- chart (f) presents uncertainty in the analysis of engine operation in relation to the vehicle’s speed on the road.

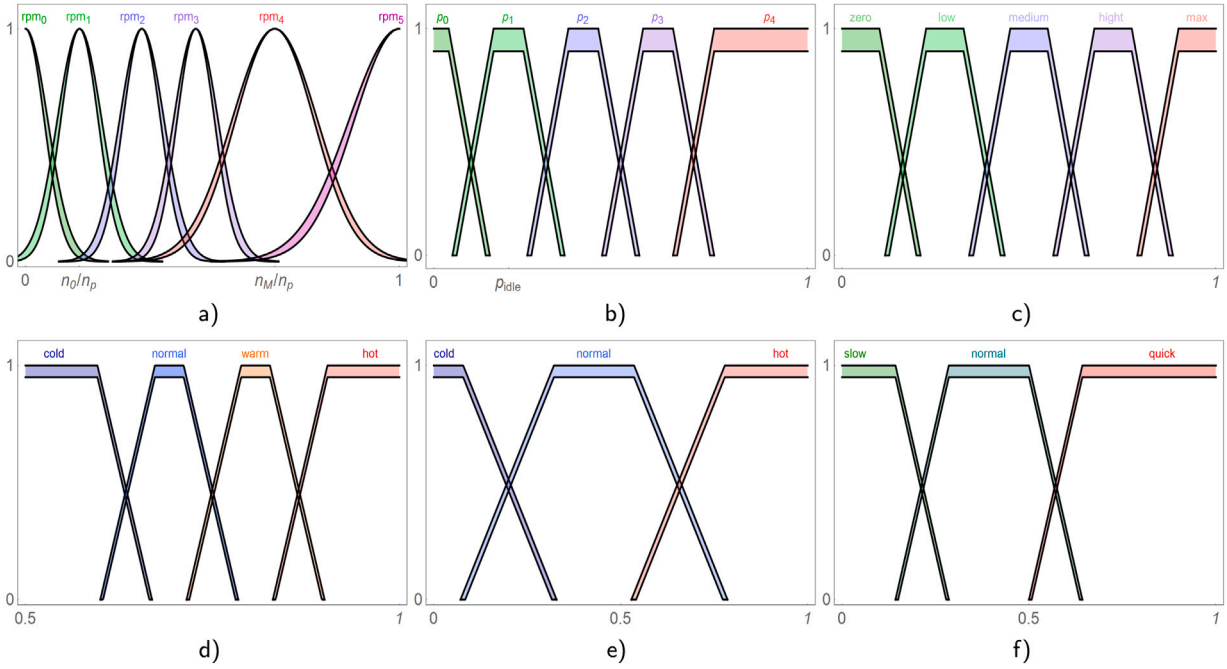


Fig. 13. Membership functions describing normalized type-2 fuzzy sets: rotational speed of the engine shaft (a), negative pressure in the engine intake manifold (b), torque developed by the engine (c), engine temperature (d), ambient temperature (e), vehicle speed (f).

Our system is a comprehensive model for the analysis of various vehicle systems. The drive system is analyzed using temperature and vacuum sensors, which allows us to control the consumption of the cooling system. During the tests, we determined a model for analyzing the operation of the silkscreen (drive system) and the suspension (depreciation) system. The test results show that both systems operate correctly. We chose vibration sensors (accelerometers) to analyze the vehicle's suspension system because they show vibrations and the harmonic spectrum shows how individual wheels work, as a result of which our system is able to analyze shock absorber wear in real time. The system correctness was confirmed at certified diagnostic station (see Fig. 18).

4.2. Fuzzy control system for engine torque

Developed method of determining the fuzzy value of the engine load torque is constructed for naturally aspirated gasoline engines. The parameters allowing to determine the fuzzy value are:

- engine speed - measured with the measuring system attached to the ignition module installed in the ignition device, this parameter for any engine has specific characteristic values: idle speed, revolutions corresponding to the maximum torque and revolutions corresponding to the maximum engine power,
- negative pressure in the engine intake manifold - measured at the output of the absolute pressure sensor (MAP sensor), this parameter takes values from the range (0.1], where 1 corresponds to atmospheric pressure and 0 to vacuum, the intermediate characteristic value is the pressure for revolutions idling.

Each of these two quantities has been fuzzified and fuzzy rules have been defined to determine the motor load moment. Due to the fact that for different naturally aspirated gasoline engines, the characteristic points of rotational speed may be different and their proportions are similar, this parameter must be standardized for each car. Therefore the driver must enter the rotational speed for: idling, maximum torque and maximum engine power. These parameters are common for a given type of engine. Normalization is performed down to the [0.1] range.

Table 1
Expert knowledge for motor torque load control.

		Crankshaft rotation per minute				
		rpm_1	rpm_2	rpm_3	rpm_4	rpm_5
Pressure	p_1	*zero*	*zero*	*zero*	*low*	*zero*
	p_2	*zero*	*zero*	*low*	*medium*	*low*
	p_3	*zero*	*low*	*medium*	*height*	*medium*
	p_4	*zero*	*low*	*height*	*max*	*height*

Gaussian functions were used to blur the rotational speed of the motor shaft. Normalized membership functions describing the rotational speed of the motor shaft are shown in Fig. 13a). Due to the fact that the vacuum values in the engine intake manifold are in the range of [0.1], this parameter does not need to be normalized. The blur was done using the trapezoidal membership functions. The fuzzy sets describing this parameter are presented in Fig. 13b). Based on a set of rules:

IF rpm is R **AND** pressure is P **THEN** moment is M

the knowledge base is included in Table 1, in which the sets row for p_0 and the column for rpm_0 are omitted, because in both cases the answer is *zero*.

Examples of engine torque determination based on rotational speed and vacuum in the intake manifold based on the proposed type-2 fuzzy system are shown in Fig. 14, where the trace of the resulting fuzzy set is marked in gray, and the lines of black color within the trace indicate the course of the sharpening algorithm, vertical the line shows the sharpened value of the motor torque.

4.3. Fuzzy control for vehicle's driveline

The primary diagnostic system will determine the condition of the vehicle's driveline based on a fuzzy values:

- moment M, fuzzification is shown in Fig. 13c),
- engine temperature item T_e, fuzzification is shown in Fig. 13d),
- ambient temperature item T_a, fuzzification is shown in Fig. 13e),
- vehicle speed item V, where the fuzzification is shown in Fig. 13 (e),

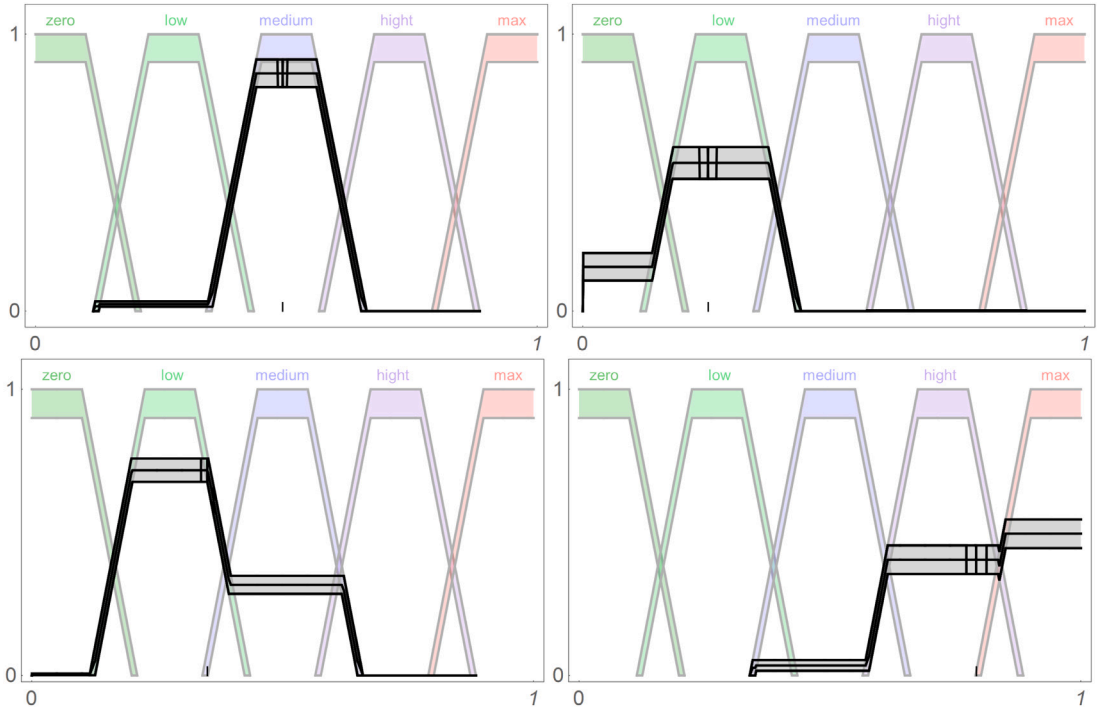


Fig. 14. Examples of determining the torque value based on rotational speed of the engine shaft and the vacuum in the intake manifold for the created fuzzy system.

Table 2
Expert knowledge for developed car diagnostic system in case of engine temperature value $T_e = warm$.

$T_e = warm$		Moment											
		zero	low	medium			high			max			
Velocity	low	*prob. bad*	*prob. bad*	Ambient temp.			Ambient temp.			*fine*			
				cold	normal	hot	cold	normal	hot				
				prob. bad	*prob. fine*	*fine*	*prob. bad*	*prob. fine*	*fine*				
	normal	*bad*	*bad*	*prob. bad*			*prob. fine*			*prob. fine*			
	quick	*bad*	*bad*	Torque converter clutch			Torque converter clutch			Torque converter clutch			
				off	on		off	on		off	on		
				prob. bad	*prob. fine*		*prob. bad*	*prob. fine*		*prob. bad*	*prob. fine*		

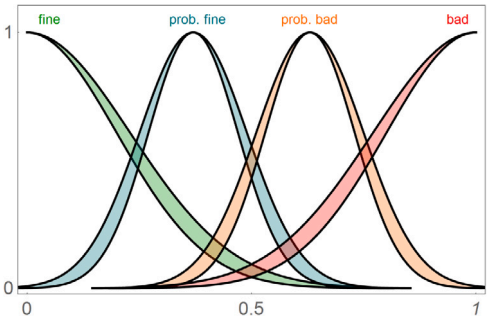


Fig. 15. Type-2 fuzzy sets describing primary diagnostic system membership functions to control condition of the vehicle drive system.

- state of torque converter clutch, as a singleton with two ON/OFF values.

The primary diagnostic system will detect possible excessive overheating of the engine under certain operation conditions. The membership functions describing the response of the diagnostic system are shown in Fig. 15. Due to the fact that the elements of the drive system have a certain thermal inertia, the system is based on the averaged values of the tested parameters in a fixed time - 5 min. We will use a moving average to determine the values of the input parameters. Based on the expert knowledge fuzzy control system was created using the following set of rules:

1. **IF** engine temperature is *cold* **THEN** machine condition is *probably bad*
2. **IF** engine temperature is *normal* **THEN** machine condition is *fine*
3. **IF** engine temperature is *warm* **AND** moment M **AND** velocity is V **AND** ambient temperature is T_a **AND** TCC state is S **THEN** machine condition is C
4. **IF** engine temperature is *hot* **THEN** machine condition is *bad*

The rule 3) is described in a general way, where the knowledge is placed in Table 2. In the table, in two cells for velocity *slow* and moment *medium* or *high* the answer depends on the ambient temperature. This means that in other cases the ambient temperature has no influence on the form of the rules. For the measurement of the temperatures in and out of the radiator (T_2, T_3) and the temperature of the gearbox oil (T_6, T_7), fuzzy rules have been defined in a similar way, allowing the detection of insufficient cooling system efficiency. In this way, the system detected contamination of the air conditioning radiator, which reduced the efficiency of the engine radiator by restricting the air flow. Fig. 16 shows four various cases of application of the created fuzzy system.

4.4. Carbon dioxide level control

We understand conditions in the vehicle cabin as comfortable when the temperature is adjusted to the users' preferences and when the

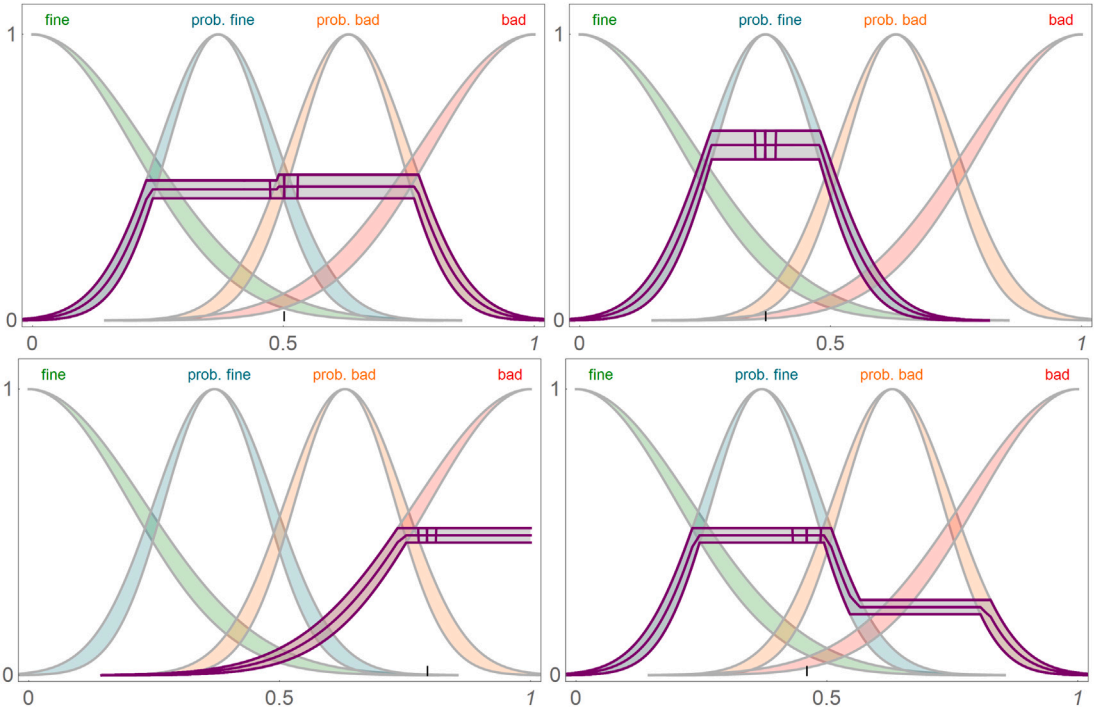


Fig. 16. Sample system responses for four different input states.

cabin is properly ventilated. In case of continuous ventilation of the cabin, the comfort of CO_2 concentration is ensured, but if the aim is to ensure thermal comfort while reducing energy consumption, ventilation is limited. When the ambient temperature is significantly higher than set by the driver, it is less energy-consuming to maintain the set cabin temperature in a closed circuit than cooling the air supplied from the outside. Due to the fact that the proposed system applies to classic cars that were not equipped with such air control system, the proposed solution uses a CO_2 sensor that periodically checks the concentration of carbon dioxide in the atmosphere of the vehicle cabin. The schematic diagram of the measuring device is presented in Fig. 10 (additionally, this system measures the temperature in the car cabin). In the event of exceeding the limit of CO_2 concentration, the system notifies the vehicle driver of the need to supply air to the car cabin. In case of solutions with electrically controlled vents, the system can be expanded so that for an excessive concentration of CO_2 it will automatically open the supply of fresh air and closes it after reaching a set acceptable concentration. A proposal of such an executive system is presented in Fig. 11. In the case of cars with mechanically regulated ventilation, the system will inform the driver about the need to ventilate the vehicle. In the same case, the operation of the system depends on whether the air conditioning is on. Then, when the CO_2 concentration is correct, the system reports the possibility of closing ventilation in order to reduce energy consumption for air cooling.

4.5. Drive transmission components and suspension system analytical control

During drive microcontroller records 1500 values from each of the channels per 1 s. Data is sent to the ESP8266-01 module, where it is buffered in memory. There are 5 one-second recording blocks saved in the ESP8266-01 module memory. In addition, this system determines the minimum and maximum values for each of the channels within 1 s. Data received in this way is sent to the central unit, which monitors it in real-time. In a situation when the second reading will indicate possible anomalies in vibrations, the central unit will send an information to the ESP8266-01 module to provide all buffered measured values. The measurement covers time interval from 2 to 3 s before triggering the measurement and is 5 s long. This operation of the system results from the fact that the central unit reaction is based on the largest and smallest values from the last seconds of the measurement. The transferred complete data of 5 s of measurement in three axes from each wheel is transformed into the frequency domain using a Fourier transform.

4.5.1. Vibration frequency analysis

The Fourier transform returns frequency spectrum for a signal specified in the time domain. This spectrum allows to determine which harmonics and with what energy are contained in the tested signal. Let $f(t) \in L^1(\mathbf{R})$ be the space of integrable functions on $\mathbf{R}$. Fourier transform for the function $f(t) \in L^1(\mathbf{R})$ is given by the formula:

$$\hat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-2\pi i (x\xi)} dx. \tag{10}$$

In case of signals in the discrete form we are dealing with the discrete form of the Fourier transform. Let $\mathbf{a} = (a_0, a_1, a_1, \ldots, a_{N-1})$ be the real-valued signal $a_i \in \mathbf{R}$, then the sequence of harmonics $\mathbf{A} = (A_0, A_1, \ldots, A_{N-1})$ where $A_j \in \mathbf{Z}$ is given by the formula:

$$A_j = \sum_{k=0}^{N-1} a_k \cdot w_N^{-jk}, \quad j = 0, 1, \ldots, N-1, \tag{11}$$

where:

$$w_N = e^{i\frac{2\pi}{N}},$$

and i is the imaginary unit, a_k is the sample value, N is the number of samples, j is the number of the harmonic we call the discrete Fourier transform. The matrix form of the discrete Fourier transform is defined as follows:

$$A = M \cdot a \tag{12}$$

where:

$$\mathbf{a} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} A_0 \\ A_1 \\ \vdots \\ A_{N-1} \end{bmatrix},$$



Fig. 17. Mobile application implemented for the car radio system. We can see how the system operates in the car, and how it was implemented using Android studio on pc machine.

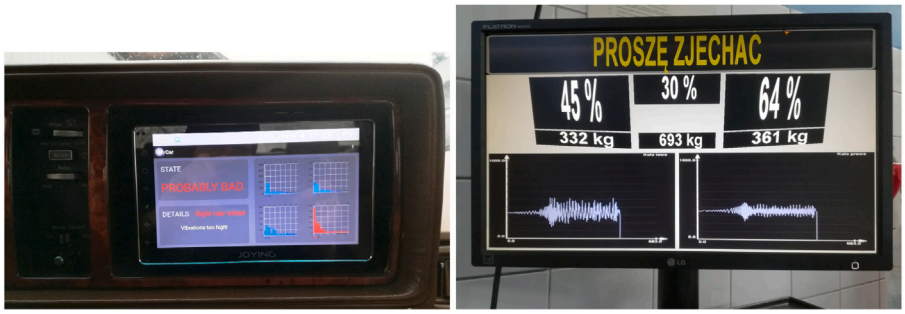


Fig. 18. A photo of a potential defect alert in developed AIoT system (on the left) and a photo from a diagnostic station showing the wear of the shock absorber (on the right), what confirms correct diagnose from our system.

$$\mathbf{M} = \begin{bmatrix} w_N^{-0\cdot 0} & w_N^{-1\cdot 0} & \dots & w_N^{-(N-1)\cdot 0} \\ w_N^{-0\cdot 1} & w_N^{-1\cdot 1} & \dots & w_N^{-(N-1)\cdot 1} \\ \vdots & \vdots & \ddots & \vdots \\ w_N^{-0\cdot (N-1)} & w_N^{-1\cdot (N-1)} & \dots & w_N^{-(N-1)\cdot (N-1)} \end{bmatrix}.$$

The equations of the car vibration model were constructed based on the Fourier transform, which allows for modeling functions in a variable time horizon. Since measurements from sensors installed in the car give readings in the form of a time series, we used a discrete Fourier transform model in our calculations.

5. Sample work and drive analysis

Fig. 17 shows pictures visualizing the implementation stage and the mobile application running in the vehicle. We have tested our AIoT system in the long run with several situations recorded when the system indicated a potential failure of the drive system or suspension module. The first example we want to present is for a rear shock that showed less damping efficiency. Figs. 19 and 20 present raw data from accelerometers recorded in the perspective of 5 s (middle row of figures), the transmission of which to the central unit was caused by anomalies



Fig. 19. Recordings illustrating the defect of the shock absorber, where the charts show measurements and analysis for the next car wheels. In the first row we have seconds data (blue line), moving average (orange line) and system operation threshold (red line). In the second row - raw data from accelerometers, in the third - the spectrum of harmonics.

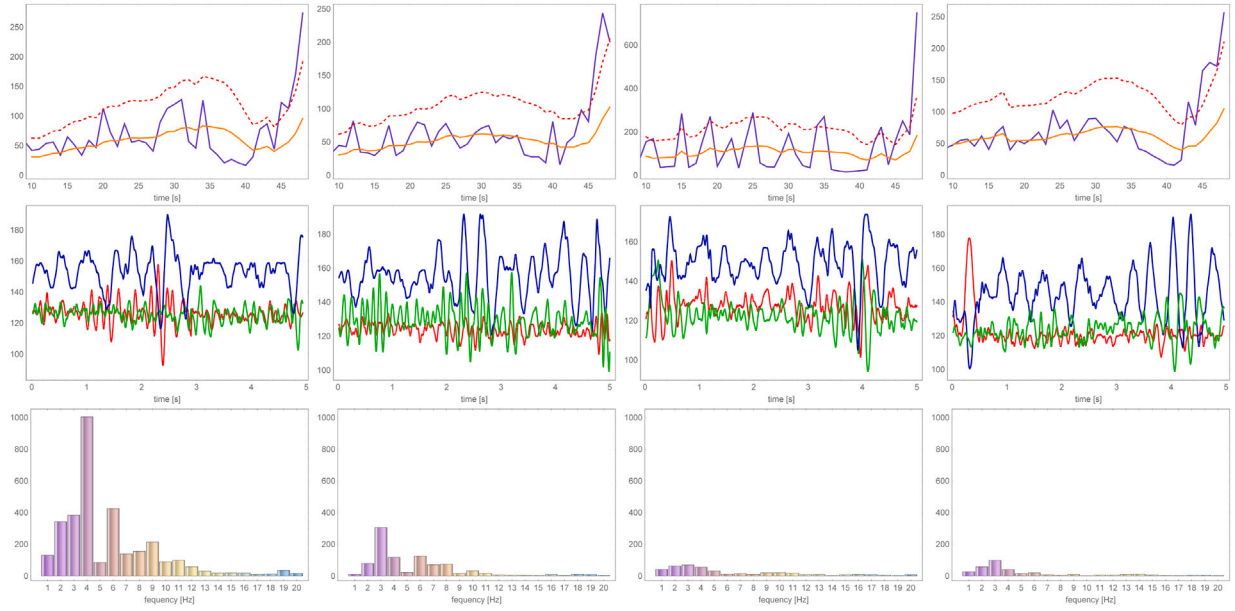


Fig. 20. Recordings illustrating the brake disc defect, where the charts show measurements and analysis for the next car wheels. In the first row we have second data (blue line), moving average (orange line) and system operation threshold (red line). In the second row - raw data from accelerometers, in the third - harmonics spectrum.

in the second measurement. Second row measurements with a moving average and its threshold are shown in the first row of figures. The third row shows the spectrum of harmonics obtained on the basis of Fourier transform analysis. These spectra suggest that in the fourth circle (4th column) vibrations of a significantly greater amplitude and thus also energy appeared. Fig. 19 presents defect of absorber, while Fig. 20 presents defect of brake disc. For the above potential failure, the system reported announcement presented in Fig. 18. A visit to the diagnostic station, after detecting potential fault causing greater vibrations on one of the wheels, has proved correct work of our system. Tests showed that the damping of one rear shock absorber was significantly lower, as shown in Fig. 18.

Temperatures in the engine compartment may vary from approximately −30 to 120 Celsius degrees. In addition, sensors installed at the engine must be resistant to vibrations and contamination with operational and environmental agents (dust, oil, operating fluids). Additionally, the entire measurement system must be resistant to electromagnetic interference resulting from, for example, the operation of the ignition system. The proposed algorithm must take into account all these potential cases and be resistant to them. In some cases of real-time system operation, certain information may be missing. Lack of information in the presented algorithm cannot cause its suspension or other undesirable effects.

6. Conclusions

As part of the research, temperatures were monitored, thanks to which it was possible to detect the decreasing parameters of the cooling

Table 3
Comparison of real-time systems.

Authors	Quality measure	Methodology	Type of object
Devapriya et al. (2016)	no provideed	Threshold-based and standard deviation	vehicle
Mikko Perttunen and Riekki (2011)	accuracy more than 80%	machine learning – SVM	car or train
Sarah Biswal (2023)	accuracy more than 75%	machine learning – SVM	car
Bhoraskar et al. (2012)	no data	machine learning – SVM	vehicle
Yi et al. (2015)	no data	threshold-based – standard deviation of sensor data	car
Singh et al. (2017)	accuracy 88.66%	DTW Technique – dynamic time warping	car
Mednis et al. (2011)	accuracy 68%–90%	Threshold-based Z-THRESH, Z-DIFF, STDEV-Z	car
Wang et al. (2018)	error 6%	signal transform – Daubechies wavelet transform	car

system over time. To detect the decreasing capabilities of the cooling system, it was necessary to measure other quantities of the drive system. These measurements and advanced analysis of the results made it possible to protect the drive system against the effects of overheating and the resulting failure. The next set of measurements were vibration measurements in the vehicle's suspension system. The set of these measurements and their analysis, in this case using the Furier transform, allowed for the detection of the deteriorating condition of one shock absorber. The reduced damping parameters of the shock absorber were additionally confirmed by tests carried out at a diagnostic station. The proposed method of additional on-board diagnostics worked well in the tested vehicle. Modern digital communication and computing capabilities significantly exceed the capabilities that existed at the time of production of the diagnosed vehicles. In Table 3 are presented other approaches to compose a diagnostic system for car or vehicle. Results show that our model composed of type-2 fuzzy control approach is more flexible and easier to adjust to different input types.

The aim of the research was to create an AIoT system, which, thanks to the non-invasive arrangement of sensors, will allow using Artificial Intelligence and mathematical apparatus as a complex control systems to detect symptoms that indicate the progressive wear of certain vehicle components. Long-term tests in research car made it possible to detect a few cases of wear of components in good time before the occurrence of a perceptible failure. The proposed system in its current form can be used without any adaptation in cars equipped with an engine controlled by a pressure sensor in the intake manifold. Of course, the system can be adapted to other solutions, but then the analysis would have to be repeated to position fuzzy system. The vibration detection system can also be extended and applied to other vehicle components. Observations of the system operation in a research car show that the system is useful and allows for earlier detection of a deteriorating condition of the monitored elements.

In future research, we plan to test our system on other oldtimer vehicles. A valuable extension of our system would also be automatic adaptation of the system to a given vehicle.

CRediT authorship contribution statement

Adam Zielonka: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andrzej Sikora:** Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Marcin Woźniak:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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